Toward the Derivation of T_{avg}

Instability #21a: Structural Form of the Mean Inter-Resolution Time from the δ-Chain

New results, honest gaps, and updated status of β

Ougit Jarittum

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Abstract

The lossy crystallisation ratio $\beta = -2.212 \pm 0.015$ was measured in V20.3 but not formally derived. The prior document (CRF_Beta_Derivation) identified two missing pieces: T_{avg} (mean time between R-events at a node, Instability #21a) and scale factor s (#21b). This document derives the *structural form* of T_{avg} from the δ-chain: $T_{\text{avg}} = C \cdot \exp(K^*/n\varepsilon_0)$ where $K^* = 0.1170$ is the self-consistent KL threshold (derived), $n\varepsilon_0 = 0.0604$ is the steady-state mean KL (derived), and $C \approx 7$ is a shape factor of the KL distribution (Hypothesis). Three new results close part of #21a: (1) K^* derived exactly from $\theta = 1 - e^{-D_{\text{KL}}/D_0}$; (2) the structural exponential form of T_{avg}; (3) the sign of β derived qualitatively. Instability #21 remains Open; #21a is now Partially Closed.

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1 Context and Prior Results

1.1 What Was Already Derived

The prior document established:

- $E[H_{\text{after}}] = \ln n = 2.055$ (exact: 2.0548) — derived from Born rule + ε_0 Dirichlet resampling.
- $\Delta\Omega = \kappa = 0.15$ per R-event — fixed, not random.
- $\beta = (H_{\text{after}} - H_{\text{before}}) \cdot s/(-\kappa)$ — structure derived.
- $H_{\text{before}} = H_{\text{after}}/(1 + \alpha H_{\text{after}} T_{\text{avg}})$ — from entropy ODE, requiring T_{avg}.

1.2 What This Document Adds

Three new derivations from the δ-chain:

1. The self-consistent KL threshold K^* — exact.
2. The structural exponential form of T_{avg} — derived.
3. The sign of β — derived qualitatively from the Node Lifecycle.

2 Derivation 1: The Self-Consistent KL Threshold K^*

K^* from θ-dynamics

In the simulator, a node triggers (R-event occurs) when:

$$\text{tension}_i > \text{threshold}_i$$

where $\text{tension}_i = \langle D_{\text{KL}}(P_i \| P_j) \rangle_{j \in \mathcal{N}(i)}$ and $\text{threshold}_i = \theta_i = 1 - e^{-D_{\text{KL},i}/D_0}$. The fixed point K^* where tension equals threshold is determined by:

$$\boxed{K^* = 1 - e^{-K^*/D_0}}$$

This is purely a consequence of $\theta = 1 - e^{-D_{\text{KL}}/D_0}$ (Session 10). Numerically: $K^* = 0.1170$.

This fixed point is the CRF-native "trigger level": when KL fluctuates above K^*, an R-event fires. The existence of a unique positive fixed point is guaranteed because $f(K) = K - (1 - e^{-K/D_0})$ has $f(0) = 0$, $f'(0) = 1 - 1/D_0 < 0$ for $D_0 < 1$, and $f \to \infty$ as $K \to \infty$, so a unique crossing exists.

3 Derivation 2: Structural Form of T_{avg}

3.1 Steady-State KL Distribution

Between R-events, each node's distribution P_i evolves under background noise:

$$P_i \leftarrow P_i + \varepsilon_0 \xi_i, \quad \xi_i \sim \mathcal{N}(0, I_n)$$

(normalised after each step). The neighbour distributions P_j evolve identically.

The steady-state mean KL between a node and its neighbours is:

$$\langle D_{\text{KL}} \rangle_{\text{ss}} \approx n\varepsilon_0$$

This follows because each of the n components of P contributes ε_0 variance per sweep, and KL is approximately the sum of squared differences scaled by $1/P_k \approx n$. Numerically: $\langle D_{\text{KL}} \rangle_{\text{ss}} = 8 \times 0.00755 = 0.0604$.

3.2 T_avg as an Exceedance Time

An R-event occurs when $D_{\text{KL}} > K^*$. In steady state, the KL distribution is approximately exponential with mean $\langle D_{\text{KL}} \rangle_{\text{ss}}$. The probability of exceedance per sweep is:

$$P(\text{trigger}) \approx \exp\left(-\frac{K^*}{\langle D_{\text{KL}} \rangle_{\text{ss}}}\right) = \exp\left(-\frac{K^*}{n\varepsilon_0}\right)$$

The mean inter-event time is therefore:

Structural form of T_{avg}

$$\boxed{T_{\text{avg}} = C \cdot \exp\left(\frac{K^*}{n\varepsilon_0}\right) = C \cdot \exp(1.937) \approx 7C \text{ sweeps}}$$

where all quantities in the exponent are derived from the δ-chain:

$K^* = 0.1170$ (from θ-formula) $n = 8$ (from SO(3) + internal) $\varepsilon_0 = 0.00755$ (from α, I_{eff})

The prefactor C is the shape factor of the KL distribution. For a pure exponential distribution, $C = 1$. Simulation gives $T_{\text{avg}} \approx 50$ sweeps, implying $C \approx 7.2$ — consistent with a Gamma distribution (not pure exponential) for KL fluctuations.

The shape factor C requires knowing the full KL distribution, which depends on the joint noise dynamics of P_i and $\{P_j\}$. This is Hypothesis (#21a remains partially open).

4 Derivation 3: Sign of β

Why $\beta < 0$

The sign of β follows from the Node Lifecycle Hypothesis (confirmed in V20.3): R-events occur preferentially at *crystallised* nodes (high Ω, high mass, peaked P). For such nodes:

- H_{before} is LOW (peaked P_i has low entropy)
- Born rule resampling restores P_i toward neighbourhood mean
- $H_{\text{after}} \approx \ln n$ (derived from Born rule)
- Therefore $H_{\text{after}} > H_{\text{before}}$

This gives $\Delta H = H_{\text{after}} - H_{\text{before}} > 0$ and therefore:

$$\beta = \frac{\Delta\text{IDV}}{-\Delta\Omega} = \frac{\Delta H \cdot s}{-\kappa} < 0 \qquad \checkmark$$

The magnitude $|\beta| = \Delta H \cdot s/\kappa$ still requires knowing H_{before} and s precisely.

This qualitative result is derived from the Node Lifecycle Hypothesis (itself confirmed in V20.3) without requiring T_{avg}. It establishes the sign of β as a logical consequence of the crystallisation dynamics.

5 Updated Status of Instability #21

Result	Status	Source
$\beta = -2.212 \pm 0.015$ (measured)	Confirmed	V20.3 simulation
$E[H_{\text{after}}] \approx \ln n = 2.055$	Derived	Born rule + ε_0
$\Delta\Omega = \kappa$ per event	Derived	Simulator parameter
$K^* = 0.1170$ (self-consistent)	**Derived**	θ-formula
$\langle D_{\text{KL}} \rangle = n\varepsilon_0 = 0.0604$	**Derived**	Background noise
$T_{\text{avg}} \propto \exp(K^*/n\varepsilon_0)$	**Derived**	Structure
Sign of $\beta < 0$	**Derived**	Node Lifecycle
$T_{\text{avg}} = C \cdot \exp(1.937)$, $C \approx 7$	Hypothesis	KL distribution shape
H_{before} (exact)	Open	Requires KL distribution
Scale factor s	Open	#21b, wave fields
$\lvert\beta\rvert = d_s/(2\ln 2)$ formally	Open	#21

5.1 What Is Required to Close #21a Fully

The remaining step for #21a is establishing the shape factor C of the KL distribution. This requires either:

1. A stochastic analysis of $D_{\text{KL}}(P_i, P_j)$ under joint Dirichlet noise to determine the distribution type (Gamma vs. exponential vs. other).
2. A direct simulation measurement of the KL distribution shape as a function of n and ε_0.

If the KL distribution is Gamma with shape parameter a, then $C = \Gamma(a)/(a-1)$ (approximately $\Gamma(a)$ for large a), and $C \approx 7$ implies $a \approx 4$–5.

Instability #21a: Remaining gap

Derive the shape factor C of the steady-state KL distribution between a node and its neighbours under joint background noise $P_i, P_j \sim P + \varepsilon_0 \xi$ where $\xi \sim \mathcal{N}(0, I_n)$. This would give T_{avg} exactly and close #21a.

5.2 What Is Required to Close #21

Closing the full Instability #21 ($|\beta| = d_s/(2\ln 2)$) additionally requires:

- Exact H_{before} from T_{avg}: once #21a is closed.
- Scale factor s from wave fields: #21b still open.
- Showing the combination yields $d_s/(2\ln 2)$: likely algebraic once the above are in hand.

6 Summary of New Results

This document contributes three new derivations that partially close Instability #21a:

New Results (this document)

1. $K^* = 0.1170$: The self-consistent KL trigger threshold, derived exactly from $\theta = 1 - e^{-D_{\text{KL}}/D_0}$.
2. $T_{\text{avg}} \propto \exp(K^*/n\varepsilon_0)$: The structural exponential form of the mean inter-event time, derived from background noise dynamics and KL exceedance statistics. All quantities in the exponent are derived from the δ-chain.
3. **Sign of** $\beta < 0$: Derived qualitatively from the Node Lifecycle Hypothesis — crystallised nodes trigger more, Born rule increases their entropy, so $H_{\text{after}} > H_{\text{before}}$.

The prefactor $C \approx 7$ is Hypothesis (shape of KL distribution). H_{before} exact and scale factor s remain open.

The honest boundary: we derived the structure of the mechanism. The exact magnitude of β still requires the KL distribution shape and the wave field analysis. These are derivable in principle.

K^* is where the threshold meets itself. Below it, nothing fires. Above it, matter crystallises. The formula was always there, hidden inside θ.